

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Examination December 2021 (2019 scheme)

Course Code: CST301**Course Name: FORMAL LANGUAGES AND AUTOMATA THEORY**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- | | | |
|----|--|---|
| 1 | Draw the state transition diagram showing a DFA for recognizing the language L over the alphabet set $\Sigma = \{a, b\}$:

$L = \{x \mid x \in \Sigma^* \text{ and the number of } a \text{ in } x \text{ is divisible by 2 or 3}\}.$ | 3 |
| 2 | Write a Regular Grammar G for the language: $L = \{0^n 1^m : n, m \geq 1\}$ | 3 |
| 3 | Construct an ϵ -NFA for the regular expression $(a+b)^*ab(a+b)^*$ | 3 |
| 4 | Using homomorphism on Regular Languages, Prove that the language $L = \{a^n b^n c^{2n} \mid n \geq 0\}$ is not regular. Given that the language $\{a^n b^n : n \geq 1\}$ is not regular. | 3 |
| 5 | State Myhill-Nerode Theorem. | 3 |
| 6 | Write a Context-Free Grammar for the language $L = \{wcw^r \mid w \in \{a,b\}^*\}$, w^r represents the reverse of w. | 3 |
| 7 | Write the transition functions of PDA with acceptance by Final State for the language $L = \{a^n b^n : n \geq 0\}$. | 3 |
| 8 | State Pumping Lemma for Context Free Languages. | 3 |
| 9 | Write the formal definition of Context Sensitive Grammar and write the CSG for the language $L = \{a^n b^n c^n \mid n \geq 1\}$. | 3 |
| 10 | Explain Chomsky hierarchy of languages. | 3 |

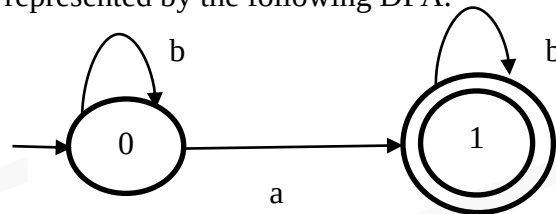
PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

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|----|---|---|
| 11 | a) Draw the state-transition diagram showing a DFA for recognizing the language: $L = \{x \in \{a,b\}^* \mid \text{every block of five consecutive symbols in } x \text{ contains two consecutive } a\text{'s}\}$ | 6 |
|----|---|---|

- b) Draw the state-transition diagram showing an NFA N for the following language L . Obtain the DFA D equivalent to N by applying the subset construction algorithm. $L = \{x \in \{a, b\}^* \mid x \text{ contains 'bab' as a substring}\}$ 8
- 12 a) Define Regular Grammar and write Regular Grammar G for the following language : $L = \{x \in \{a, b\}^* \mid x \text{ does not ends with 'bb' }\}$ 7
- b) Obtain the DFA over the alphabet set $\Sigma = \{a, b\}$, equivalent to the regular grammar G with start symbol S and productions: $S \rightarrow aA \mid bS$, $A \rightarrow aB \mid bS \mid a$ and $B \rightarrow aB \mid bS \mid a$ 7

Module -2

- 13 a) State and explain any three closure properties of Regular Languages. 6
- b) Find the equivalent Regular Expression using Kleene's construction for the language represented by the following DFA. 8



- 14 a) Using pumping lemma for Regular Languages, prove that the language $L = \{0^n \mid n \text{ is a perfect square}\}$ is not Regular. 7
- b) Obtain the minimum state DFA for the following DFA. 7

	a	b
→ 0	1	2
1	4	5
2	0	3
3	5	2
4	1	0
5	4	3

Module -3

- 15 a) Show the equivalence classes of Canonical Myhill-Nerode relation for the language of binary string which starts with 1 and ends with 0. 7
- b) Consider the following productions: 7
- $S \rightarrow aB \mid bA$
- $A \rightarrow aS \mid bAA \mid a$

$B \rightarrow bS \mid aBB \mid b$

For the string 'baaabbba' find

- i) The leftmost derivation
 - ii) The rightmost derivation
 - iii) The parse tree
- 16 a) Construct the Grammars in Chomsky Normal Form generating the set of all strings over $\{a,b\}$ consisting of equal number of a's and b's. 7
- b) Find the Greibach Normal Form for the following Context Free Grammar 7
- $S \rightarrow XA \mid BB, B \rightarrow b \mid SB, X \rightarrow b, A \rightarrow a$

Module -4

- 17 a) Design a PDA for the language $L = \{ww^r \mid w \in \{a,b\}^*\}$. Also illustrate the computation of the PDA on the string 'aabbbaa'. 7
- b) Construct a CFG to generate $L(M)$ where $M = (\{p, q\}, \{0, 1\}, \{X, Z_0\}, \delta, q, Z_0, \emptyset)$ where δ is defined as follows: 7
- $\delta(q, 0, Z_0) = (q, XZ_0)$
- $\delta(q, 0, X) = (q, XX)$
- $\delta(q, 1, X) = (p, \epsilon)$
- $\delta(p, 1, X) = (p, \epsilon)$
- $\delta(p, \epsilon, X) = (p, \epsilon)$
- $\delta(p, \epsilon, Z_0) = (p, \epsilon)$
- 18 a) Using pumping lemma for Context free languages, prove that the language $L = \{a^n b^n c^n \mid n \geq 1\}$. 7
- b) Prove that CFLs are closed under Union, Concatenation and Homomorphism. 7

Module -5

- 19 a) Design Linear Bounded Automata for the language $L = \{a^n b^n c^n \mid n \geq 1\}$. 7
- b) Design a Turing Machine for the language $L = \{a^n b^{2n} \mid n \geq 1\}$. Illustrate the computation of TM on the input 'aaabbbbb'. 7
- 20 a) Design a Turing Machine to obtain the product of two natural numbers a and b both represented in unary on the alphabet 0. For example, number 5 is represented as 00000 ie 0^5 . Assume that initially the input tape contains $0^a 10^b$ and Turing machine should halt with 0^{a*b} as the tape content. 7
- b) Prove that 'Turing Machine halting problem' is undecidable. 7
